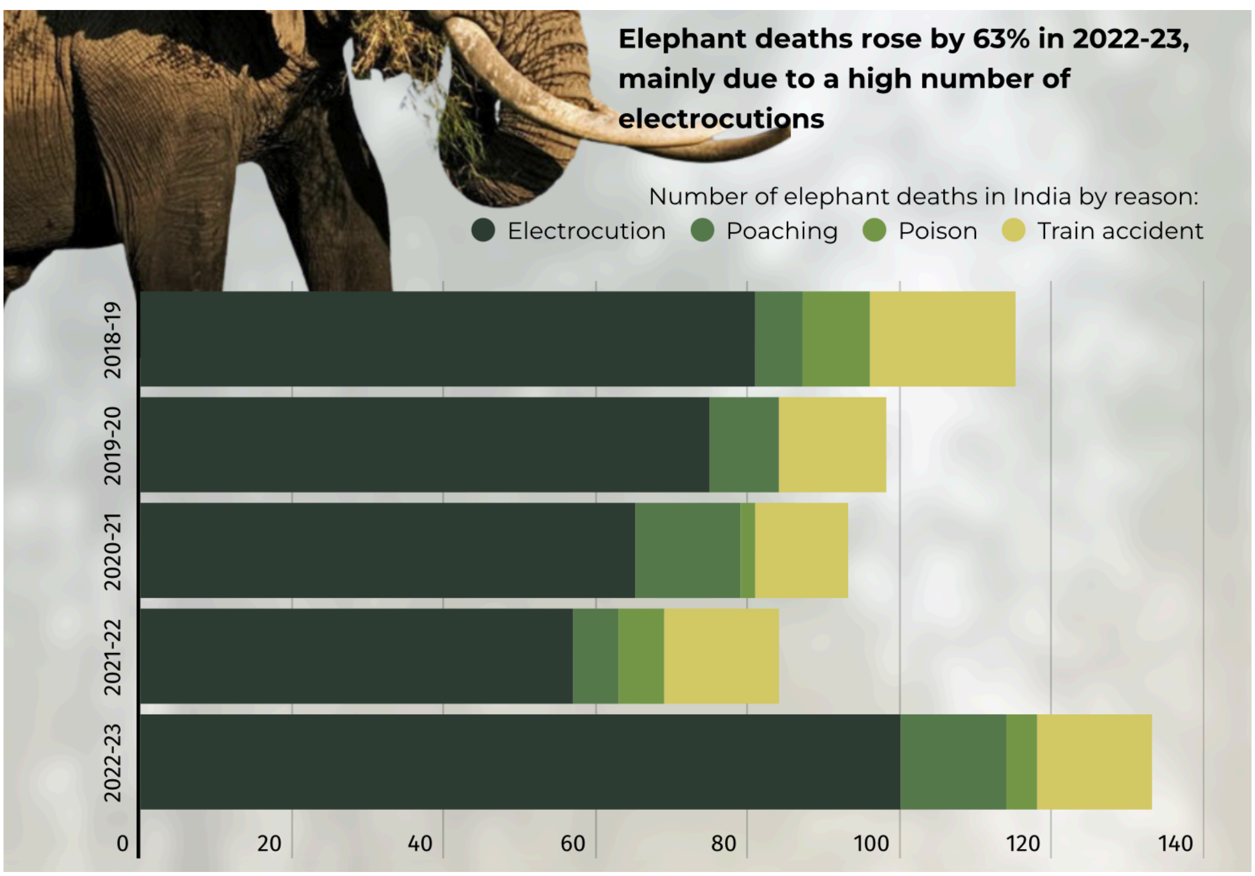
To: Shri Ashwini Vaishnaw, Hon'ble Minister for Railways; Information & Broadcasting; Electronics & Information Technology [1]
Date: February 4, 2025 [2]
CC: Shri Bhupender Yadav, Hon'ble Minister of Environment, Forest and Climate Change, Government of India
Shri Kirtivardhan Singh, Minister of State (MoEFCC)
Subject: Addressing Implementation Barriers in Expansion of Elephant Intrusion Detection Systems (EIDS) for Train-Wildlife Collision Prevention

---

Executive Summary

Official data for 2022-23 shows that death from train collisions is the second-highest cause of unnatural elephant deaths after accidental electrocution (Figure 1). More than 200 elephants were killed in train collisions in the past 10 years.[3] In June 2015, the Ministry of Railways (MoR) circulated the recommendations of the World Wildlife Fund-India to stop elephant deaths on railway tracks to six zonal railways.[4] Since then, even after joint surveys have been completed on 76% of the 110 identified stretches across 13 states to assess and mitigate elephant and tiger mortality on railway tracks, only 6% of the mitigation plans have been implemented.[5]
Since the implementation of AI-driven monitoring in 2023, over 600 elephants have been saved, demonstrating the system's potential to prevent fatalities if expanded further.[6]

Figure 1- Elephant Deaths(2022-23) [7]



Key Barriers

Despite promising funding[8] for AI-based tracking solutions to mitigate collision incidents, implementation has been slow due to policy misalignment at the state level, lack of coordination between ministries, and misallocation of funds.

Bureaucratic Approvals and Policy Misalignment- Slow environmental clearance processes at the state level have delayed AI system implementation, with some projects taking over a year for approval[9]. These state-level variations in clearance norms hinder uniform AI deployment across wildlife corridors.

Inter-Ministerial Coordination Gaps- No single agency is accountable for EIDS implementation, leading to conflicts between the MoR, MoEFCC, and state governments[10]. These hurdles have limited EIDS's effectiveness in preventing wildlife casualties along railway corridors.

Budget Constraints and Financial Prioritization- Railway modernization has received disproportionate funds, sidelining wildlife protection measures[11]. Underfunding has led to limited testing, slow scale-up, and a lack of maintenance for AI systems [12].

Long-Term Implications

Wildlife wreckage poses a significant risk to our motherland's ecological heritage.[13] Thus, investing in AI-driven tracking will reduce long-term costs associated with train damage, delays, emergency rescue efforts, and veterinary expenses.[14] This will also ensure the protection of unaware cattle and other animals. Since the Indian Railways has already embraced the Public-Private Partnerships (PPP) model for infrastructure development [15] , to offset costs, PPP and CSR contributions are well-suited as an alternate financing mechanism for AI-based conservation projects.[16]

Counter Arguments

Criticism- Some argue that reducing train speeds in wildlife corridors and constructing underpasses or overpasses are more practical than implementing costly AI tracking systems.

- **Rebuttal-** Speed restrictions are often ignored or difficult to enforce, and infrastructure-based solutions require long construction timelines and high land acquisition costs.[17] AI-driven monitoring is a faster, scalable, and dynamic solution complementing existing measures [18].

Criticism- Many argue that land acquisition issues and the need for extensive education in rural areas make the implementation of development projects, especially those addressing wildlife conflicts, impractical.

- **Rebuttal-** Recent experiences suggest that rural communities are not only open to learning but can also become active participants in such initiatives[19]. Therefore, the perception that land conflicts and the need for project education are insurmountable issues may overlook the potential for positive outcomes through community engagement and empowerment.

Analytical Endnotes

1. This memo is addressed to both the Hon'ble Ministers because railway-wildlife collisions require a coordinated response. The MoR is responsible for implementing AI-based monitoring systems, railway safety measures, and infrastructure upgrades to mitigate wildlife fatalities. And, the MoEFCC oversees wildlife conservation policies, habitat protection, and environmental impact assessments necessary for the success of AI-driven interventions. Without inter-ministerial collaboration, efforts to scale Gajraj AI and other Intrusion Detection System (IDS) technologies will remain fragmented, delaying critical action on wildlife protection and railway modernization
2. The current policy and environmental landscape heighten the urgency of this memo. India's commitment to the Kunming-Montreal Global Biodiversity Framework provides an opportunity for policy alignment and targeted conservation funding (Ghanekar, 2024).
3. “There’s blood on the tracks when railway lines go through forest areas,” quotes Satyanarayan, head of Wildlife SOS (Dhillon, A. (2024)
4. CAG Report No.5 of Railways (2021, section 2.1)
5. ArcGIS Dashboards, n.d.
6. Quote from Sangita Iyer, head of an Elephant Conservation Foundation (Casmitjana, 2024)
7. DTE Staff (2024), figure created by authors.
8. Ministry of Railways: Year End Review (2024, para. No Elephant Intrusion Detection System)
9. Dave M. (2024, paras. 1–7)
10. The author concluded that the delay in the state has been due to the lack of a State Environment Impact Assessment Authority (Dave M. 2024, paras. 1–7)
11. Budget constraints are often due to prioritization rather than a lack of funds (Fernbach et al., 2015). Railway expansion is prioritized over conservation, and there is limited publicly available data detailing the specific allocation of funds toward wildlife protection measures within these modernization projects. In the Press release for Indian Railways Major Investments for 2024, there is no mention of funds allocation for IDS or AI-based systems. (Ministry of Railways, Press Information Bureau n.d.)

12. Conservationists argue that the current IDS has an insufficient detection range for timely intervention. Experts recommend extending this to 500 meters for better response times (YourStory, 2024), but there has been slow progress in testing and improvements due to limited funds.
13. More than 63,000 animals had been run over by trains between 2017-18 and 2020-21(Shuchita Jha, 2023)
14. AI-driven tracking reduces long-term costs (Kaluvakuri, 2024).
15. CW Team (Railways to Adopt PPP Model for New Infrastructure Projects, 2024)
16. Public-Private Partnerships and CSR funding can bridge financial gaps. Major corporations, including Reliance, Tata, Infosys, and JSW, have begun investing in wildlife conservation through CSR initiatives. (Bhattacharya, The CSR Journal, 2025)
17. Speed reduction may be ineffective unless it is drastic (Rea et al., 2010)
18. The Gajraj AI system is built to use the existing OFC cables as sensors to identify elephant movements near railway tracks and alert locations to station masters, loco pilots and control offices (Dash, 2023).
19. Research emphasizes the significance of customary methods in mitigating conflicts between humans and wildlife and improving livelihoods (Hussain et al., 2022)

Bibliography

Ministry of Railways: Year end Review 2024. (n.d.-b). https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2088668

Ghanekar, N. (2024, November 1). With 23 national targets, India submits biodiversity protection plan to global body. *The Indian Express*. https://indianexpress.com/article/india/national-targets-india-submits-biodiversity-protection-plan-global-body-9648725/

Dhillon, A. (2024, April 9). Why are so many of India's elephants being hit by trains? *The Guardian*. https://www.theguardian.com/environment/2024/apr/08/why-are-so-many-of-india-elephants-being-hit-by-trains-aoe

Railway Board. (2021). Report No.5 of 2021 (Railways). In *Chapter 1* (p. 141).

https://cag.gov.in/uploads/download_audit_report/2021/Chapter%202-061a614d3778089.63745419.pdf

ArcGIS dashboards. (n.d.). https://bhlab.maps.arcgis.com/apps/dashboards/f51cd24fa5324067b51f7c756ba51c19

Casmitjana, J. (2024b, November 27). How AI Is Saving India's Elephants from Train Collisions - UnchainedTV. *UnchainedTV*. https://unchainedtv.com/2024/11/26/how-ai-is-saving-indias-elephants-from-train-collisions/

DTE Staff, & DTE Staff. (2024, August 31). *State of Environment in Figures 2024 on biodiversity in India*. Down to Earth. https://www.downtoearth.org.in/wildlife-biodiversity/state-of-environment-in-figures-2024-on-biodiversity-in-india

Ministry of Railways: Year end Review 2024. (n.d.). https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2088668

Fernbach, P. M., Kan, C., & Jr. Lynch, J. G. (n.d.). Squeezed: Coping with Constraint through Efficiency and Prioritization. In JOURNAL OF CONSUMER RESEARCH, Inc., *JOURNAL OF CONSUMER RESEARCH* (Vol. 41). https://doi.org/10.1086/679118

Dave, M. (2024b, September 30). Environmental clearance, not infrastructure, holds up more than a thousand projects in Gujarat. *The Secretariat*. https://thesecretariat.in/article/environmental-clearance-not-infrastructure-holds-up-more-than-a-thousand-projects-in-gujarat

Ministry of Railway. (n.d.). *Indian Railways 2024: Major Investments, Enhanced Safety, and Modernization to Drive Future Growth*. https://pib.gov.in/PressNoteDetails.aspx?NoteId=151988&ModuleId=3®=3&lang=1

Chakrapani, S. (2024, November 28). AI system seeks to help prevent train-elephant collisions, but experts raise questions. *YourStory*. https://yourstory.com/socialstory/2024/11/ai-system-prevent-train-elephant-collisions-conservationists-questions

Shuchita Jha, & Shuchita Jha. (2023b, January 9). *73 elephants, 4 lions run over by trains in 4 years along with 63,000 other animals: CAG report*. Down to Earth. https://www.downtoearth.org.in/wildlife-biodiversity/73-elephants-4-lions-run-over-by-trains-in-4-years-along-with-63-000-other-animals-cag-report-87011

Kaluvakuri, V. P. K. (2024b, June 30). *MAXIMIZE FLEET VALUE AND SAFETY WITH AI: REAL-TIME VEHICLE TRACKING, TELEMATICS AND COMPLIANCE SOLUTIONS: 10.55434/CBI.2024.10103*. https://www.caribjscitech.com/index.php/cjst/article/view/318

Railways to adopt PPP model for new infrastructure projects. (2024c, December 30). https://www.constructionworld.in/transport-infrastructure/metro-rail-and-railways-infrastructure/railways-to-adopt-ppp-model-for-new-infrastructure-projects/66910

Bhattacharya, H. T. a. A. (2025, January 29). *Top 100 Companies for CSR and Sustainability in 2024 - The CSR Journal*. The CSR Journal. https://thecsrjournal.in/top-companies-corporate-social-responsibility-csr-sustainability-2024/

Ajaz Hussain, B.S. Adhikari, S. Sathyakumar, G.S. Rawat,
Assessment of traditional techniques used by communities in Indian part of Kailash Sacred Landscape (KSL) for minimizing human-wildlife conflict, Environmental Challenges, https://www.sciencedirect.com/science/article/pii/S2667010022001044

Dash, D. K. (2023, November 29). Railways to expand Gajraj system & Kavach to prevent elephant deaths & collisions. *The Times of India*. https://timesofindia.indiatimes.com/india/railways-to-expand-gajraj-system-kavach-to-prevent-elephant-deaths-collisions/articleshow/105601670.cms

Rea, R. V., Child, K. N., & Aitken, D. A. (2010). *YOUTUBE (TM) INSIGHTS INTO MOOSE-TRAIN INTERACTIONS*. https://www.alcesjournal.org/index.php/alces/article/view/67

Sitharamaraju, K., Beerelli, S. K., Saket, A., & Sundaravalli, N. (2020). *Public-Private Partnership (PPP) in Indian Railways: models, framework, and policies*. https://www.iima.ac.in/sites/default/files/rnpfiles/4072329742020-12-05.pdf